Gravity in the Mesh Framework: Newtonian Success, a Classical Obstruction, and the Induced Einstein Metric That Resolved It

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Version 3, August 2026 -- brought current with the induced completion (derived, and matching observation: gamma = 1, 1.751 arcseconds) and the strong-field campaign of v3.4.0-v3.5.0; the classical no-go is retained as the historical theorem that forced the route. Version 2, July 2026 -- a full rebuild on external review. Version 1 grew by correction addenda until it contained two mutually contradictory papers in one file; a reader stopping early received the opposite of the final result. The complete v1 record, with its addendum trail, is archived in the corpus (rope_gravity_v1_archived.docx); the machine-verified claim registry (GRV-001 through GRV-017) is the authoritative history.

Abstract

We report the complete classical gravity investigation of the Mesh Programme. The Newtonian sector succeeds: inverse-square attraction follows from static force balance of the elastic network, with the $1/r$ potential forced by three-dimensional elastostatics and the $1/r^2$ force made topology-independent by tension-flux conservation. The Einstein weak-field sector fails, and fails as a theorem: within the defined class of local, harmonic, co-material response channels, every constructible mechanism yields an effective PPN parameter $\gamma \in [-1, 0]$ against the measured $\gamma = 1$, predicting 0.44 arcseconds of solar light deflection against the photographed 1.75. The classical response channels were therefore RULED OUT as a route to Einstein weak-field gravity — a negative theorem the corpus keeps at full strength, because it did its job: it pointed at the one door left open. That door has since been walked through. The quantum-induced completion, a conjecture when this obstruction was proved, is now DERIVED: the fluctuation-induced effective action's IR-universal remainder is Einstein–Hilbert (GRV-025, Derived), the physical one-metric dictionary makes the identification unconditional (GRV-029, Derived), the strong-field extrapolation is certified a controlled expansion (GRV-048, Derived), and the induced Newton constant carries the corpus's first derived exponent pair, G = c^3 a^2/(16 pi zeta hbar) with zeta = 1.208 (GRV-075). And the numbers follow: gamma = 1 and the 1.751-ARCSECOND solar deflection are DERIVED — first as a two-condition theorem (GRV-026), strengthened to one condition when the locus test screened the induced scalar (GRV-028), and made UNCONDITIONAL by the physical one-metric derivation (GRV-029), which promotes the full classical table: light deflection 1.751 arcseconds against the photographed 1.75, Mercury perihelion 43.0 arcseconds per century, Shapiro gamma = 1, Nordtvedt eta = 0 (GRV-002). The sector's standing result is therefore a pairing: Einstein weak-field gravity is IMPOSSIBLE for the classical response channels and DERIVED, numbers and all, for the quantum-induced one — the no-go and its resolution, both by theorem, with the resolution matching the photograph the no-go could not. The obstruction's premises remain enumerated as an explicit attack surface and have survived their first external attacks, twice emerging strengthened.

Status table (read this first)

Sub-sector	Status	Basis
Newtonian mechanism	DERIVED	Force balance -> Poisson -> 1/r (GRV-005); flux conservation -> 1/r^2, topology-independent (GRV-013)
Zero point (uniform vacuum)	DERIVED (structural)	Uniform network of any tension is flat; deviations gravitate (GRV-004)
Einstein weak field via CLASSICAL response channels	RULED OUT by no-go theorem (kept at full strength; this row is the ruled-out ROUTE, not the sector verdict – see next row)	gamma in [-1,0] vs measured 1; 0.44'' vs 1.75'' (GRV-008..013, 015, 017)
Quantum-induced completion	DERIVED, matches observation	Induced EH (GRV-025); gamma = 1 and 1.751'' deflection vs photographed 1.75'' (GRV-026, GRV-028); full table unconditional – Mercury 43.0''/cy, Shapiro, Nordtvedt (GRV-029, GRV-002); controlled extrapolation (GRV-048); G exponents (GRV-075)

How to read the table: the third and fourth rows are one story in two acts, not a contradiction. The no-go theorem eliminated every CLASSICAL route to Einstein weak-field gravity and is retained at full strength precisely because it is what forced the sector through the quantum-induced door — where gamma = 1 and the 1.751-arcsecond deflection were then derived. The framework's weak-field gravity is not falsified; its classical imitation is.

Sector classification: NO-GO PLUS RESOLUTION — Newtonian mechanism derived; classical route ruled out by theorem (historical); quantum- induced completion DERIVED at weak field WITH THE MEASURED NUMBERS (gamma = 1; 1.751 arcseconds), with the strong-field extrapolation certified controlled and the black-hole programme built on it (see the Addendum and papers/rope_blackholes.pdf).

1. The Newtonian sector: what the network derives

A mass-knot is a stressed defect; static force balance of the elastic network (div stress = −f) IS the conservation law, so the Poisson equation and the 1/r potential are forced by three-dimensional elastostatics, and the tension-flux conservation theorem makes the $1/r^2$ force law topology-independent — no branching or merging of ropes escapes it. Action-at-a-distance dissolves in pre-existing connection. The zero-point theorem resolves the vacuum problem's static half: a uniform network of any tension density is gravitationally flat; only deviations gravitate. Gaede's mechanism, formalized, is credited with the Newtonian success (Appendix A).

2. Historical: the target metric, and why matching it was not a derivation

The programme's original gravity claim exhibited a conformally-matched isotropic Schwarzschild metric reproducing $\gamma = \beta = 1$ and the 1.751'' deflection. Its correct classification, adopted here on external review: TARGET ANSATZ — it reproduces general relativity by construction and was not generated by the identified rope mechanics. Its legitimate role was to define the target the mechanics must hit. The campaign below establishes that, within the defined classical class, the mechanics cannot hit it. Nothing in this section is evidence for the framework, and the registry entry (GRV-001) carries the same reclassification.

3. The campaign, at a glance

Factor-2 reduced to one linear response constraint; EP identity derived (GRV-008)

↓ Strain conditioning rejected: gamma = -4/7 (GRV-009)

↓ Mode bath rejected after three self-corrections: gamma = -1/2 universally (GRV-010)

↓ Anisotropic route: exact tidal theorem + the range obstruction (GRV-011)

↓ Classical verdict registered at full strength (GRV-012)

↓ Channel-space hunt concluded: shadows, anchored tension, branching, threshold contact all excluded (GRV-013)

↓ Quantum-induced completion conjectured -- a separate hypothesis, not an extension (GRV-014)

↓ Reviewer's graph-structural attack absorbed: fourth response column derived; clocks decouple; sign-lock theorem; no-go strengthened (GRV-015)

↓ Premise audit: four premises reduce to two primitives; topological exception allocated to EM, triply closed (GRV-016)

↓ Kelvin loophole closed by the no-monopole lemma (GRV-017)

4. The theorems, with premises as hypotheses

PRECISION NOTES adopted on external review govern every statement below. (a) The gauge statement, correctly phrased: local constitutive energy is invariant under rigid translation and therefore depends on deformation gradients rather than absolute displacement — uniform translation is unobservable; spatially varying displacement is physically meaningful. (b) THE NO-MONOPOLE LEMMA (GRV-017), closing a real loophole: in three-dimensional elasticity a point FORCE gives displacement ~ $1/r$ (Kelvin) and strain ~ $1/r^2$; but a static, isolated defect in equilibrium exerts zero net force on the medium (else it accelerates), so the Kelvin amplitude — which IS the net force — vanishes for every admissible mass-source, and the ladder begins at the force-free order: $u \sim 1/r^2$, strain ~ $1/r^3$. (c) The sourcing-exhaustion argument operates under EXPLICIT HYPOTHESES, stated as part of the theorem rather than as mechanisms not found: H1 one independent harmonic scalar; H2 local constitutive response; H3 quasi-static transport; H4 no scale-free nonlocal kernel; H5 no independent tensor order parameter; H6 no long-range orientational phase unrelated to anchored-rope fractions. THE RESULTS: the EP identity (local wave speed invariant under arbitrary conditioning, including the tortuosity channel — co-materiality, derived not assumed); the single-γ theorem with $\gamma = 1$ iff $\sigma = l + m/2 - (3/2)\tau$; the metric-order obstruction (strain capped at tidal order $1/r^3$ by the no-monopole lemma, while the metric requires potential order $1/r$ — with the substrate containing GR's tidal tensor (2, −1, −1) exactly, one derivative too high); and sourcing exhaustion (potential-range fields are direction-blind and sign-locking; direction carriers dilute as $1/r^2$ by solid angle; tension flux is conserved through spheres). CLASS-SPECIFIC CONCLUSION, stated at its actual strength: every mechanism within the defined class of local, harmonic, co-material rope responses yields $\gamma \in [-1, 0]$. Engineered classical media with multiple coupled fields, nonlocal constitutive laws, tensor order parameters, or background flows lie outside this class and this claim.

5. The verdict

The classical response channels were ruled out as a route to Einstein weak-field gravity: their derived prediction was $\gamma = -1/2$ and 0.44 arcseconds of solar deflection against the measured $\gamma = 1$ ($|\gamma - 1| < 2.3\times10^{-5}$, Cassini) and the photographed 1.75 arcseconds. That verdict is unchanged and permanent for the defined class. What has changed since it was written is the status of the escape it named: quantum-induced metric dynamics, then the leading conjecture, is now the sector's derived completion, and it RETURNS THE MEASURED NUMBERS: gamma = 1 and 1.751 arcseconds of solar deflection, derived as a theorem (GRV-026), strengthened (GRV-028), and made unconditional with the full classical table — Mercury, Shapiro, Nordtvedt — by the one-metric derivation (GRV-029). The same photograph that hanged the classical sector at 0.44 arcseconds is, for the induced sector, a match at 1.751 against 1.75. The correct weak-field gravity of this framework is carried by the weave's fluctuations, not by its classical response, exactly as the no-go required — and it agrees with every classical test to the precision the corpus has computed.

6. The escape space, honestly enumerated

Quantum induction was the leading conjecture when this section was written — independently motivated before the verdict (the ħ-entanglement of G; the registered Sakharov route; the exact Lorentz invariance of the coupled sector) — and it has since been derived (see the Status table and Section 5); this enumeration is kept as the historical record of the escape space as it stood. It was then not the unique logical escape. Also outside the theorem's hypotheses, each substantially altering the ontology: an additional independent long-range field with the required anti-correlated response (violating H1); a genuinely nonlocal constitutive law (H4); a non-equilibrium substrate (H3); a collective tensor order parameter not reducible to local strain (H5); a long-range orientational phase (H6); or modification of interpenetrability and the local-response premise. None currently has a derivation; the quantum route is 'leading' because it alone is motivated elsewhere in the corpus rather than invented for the escape.

7. The attack surface, and its record under fire

The premises reduce to two primitives — local response with translation-invariant constitutive energy (gauge up to topology), and harmonic statics outside sources — plus hypotheses H1–H6 for sourcing exhaustion. The topological exception is real and allocated: winding holonomy is physical and IS the electromagnetic sector; gravity cannot recruit it (neutral sources give dipole-and-faster fields; Eötvös universality excludes charge-coupled free fall to $\sim 10^{-13}$; global constraints are zero-modes). Record under attack: the graph-structural challenge produced the fourth response column and a strengthened no-go; the Kelvin challenge produced the no-monopole lemma. Within the presently identified classes of classical response channels, no remaining avenue is known. Conditions that would overturn this conclusion: a classical channel evading solid-angle dilution; an anti-correlated second harmonic field arising from stated commitments; a demonstrated $1/r$ traceless conditioning at locked ratio $1/4$; an explicit induced-action calculation yielding $\gamma = 1$; or experimental contradiction of the scalar prediction where classical channels dominate.

8. Lessons

What failed: isotropic conditioning of every construction; strain; the mode bath; anchored and branching tension; threshold contact; graph tortuosity. What survived: the Newtonian mechanism; the Poisson derivation; the zero-point theorem; the exact tidal geometry; the four-response machinery. What became clearer: metric order differs from strain order by one derivative no classical channel crosses; scalar media naturally produce $\gamma \approx -1/2$, explaining a century of mechanical gravity theories that captured Newton and failed light deflection; Newton is the fixed point of conservation plus solid angle, and the same conservation forbids Einstein — a mechanism's Newtonian success and its Einsteinian failure can be the same mathematical fact. The framework-independent statement of these results is the companion paper, 'A Metric-Order Obstruction for Classical Harmonic Media.'

Appendix A: Gaede’s proposal, mapped onto this verdict

Gaede's gravity — tension in the literal ropes interconnecting all atoms, with light and gravity one mechanism — is credited with the Newtonian success: formalized, his anchored-tension picture delivers the inverse-square force by flux conservation, dissolves action-at-a-distance, and ties gravity's weakness to rope-count dilution. His non-uniform-tension claim is mechanically realizable through branching (trunk tension sums), and fails by the same conservation theorem that grants him Newton. What his account never engages is the Einstein-level phenomenology — and 1.75 arcseconds is a photographed angle, not a formalism. Under his own visualizability criterion no verdict is renderable; under this programme's quantitative standard, the verdict is Section 5. The credit and the criteria difference are both on the record.

A mechanism's Newtonian success and its Einsteinian failure can be the same mathematical fact.

Registered Claims in This Paper

Each statement below is traceable to the machine-readable registry (claims.yaml), the corpus's single source of truth. Status labels: Derived (follows from stated mechanics), Modeled (reproduces data with declared inputs), EFT-constrained (bounded by effective-theory limits), Conjecture (proposed, not derived), Open (registered unsolved problem), Failed (falsified, kept as a finding). Where a claim is code-backed, the named benchmark reruns it.

- **GRV-001 [Modeled]:** Rope effective metric is IDENTIFIED WITH isotropic Schwarzschild in the weak field [benchmark: benchmarks/reproduce_results.py]

- **GRV-002 [Derived]:** Given the matched weak-field metric [benchmark: benchmarks/reproduce_results.py]
- **GRV-003 [Modeled]:** Weak-field field equation [benchmark: benchmarks/gravity/weak_field_poisson.py]
- **GRV-004 [Modeled]:** ZERO-POINT THEOREM [benchmark: benchmarks/gravity/zero_point_and_poisson.py]
- **GRV-005 [Modeled]:** GRV-003's assumed premise DERIVED from network statics [benchmark: benchmarks/gravity/zero_point_and_poisson.py]
- **GRV-006 [Derived]:** G is NOT derivable from current commitments [benchmark: benchmarks/gravity/g_underdetermination.py]
- **GRV-007 [Modeled]:** CONJECTURE [benchmark: benchmarks/gravity/g_underdetermination.py]
- **GRV-008 [Derived]:** TENSOR STRUCTURE REDUCED [benchmark: benchmarks/gravity/tensor_structure_reduction.py]
- **GRV-009 [Failed]:** FAILED CANDIDATE [benchmark: benchmarks/gravity/tensor_structure_reduction.py]
- **GRV-010 [Failed]:** FAILED [benchmark: benchmarks/gravity/mode_bath_gamma.py]
- **GRV-011 [Modeled]:** EXECUTED [benchmark: benchmarks/gravity/anisotropic_defect_field.py]
- **GRV-012 [Open]:** STANDING ADVERSE VERDICT [benchmark: benchmarks/gravity/anisotropic_defect_field.py]
- **GRV-013 [Derived]:** THE HUNT CONCLUDED -- NO-GO [benchmark: benchmarks/gravity/locked_quadrupole_hunt.py]
- **GRV-014 [Conjecture]:** CONJECTURE [benchmark: benchmarks/gravity/quantum_completion_audit.py]
- **GRV-016 [Modeled]:** PREMISE-INDEPENDENCE AUDIT [benchmark: benchmarks/gravity/premise_independence_audit.py]
- **GRV-017 [Derived]:** NO-MONOPOLE LEMMA [benchmark: benchmarks/gravity/no_monopole_lemma.py]
- **GRV-018 [Derived]:** INTERNAL-MODE DICHOTOMY [benchmark: benchmarks/gravity/internal_mode_dichotomy.py]

Naming note (5 September 2026). This programme is now the Mesh Programme and the theory it develops the mesh framework; 'the rope framework' in earlier revisions reads 'the mesh framework'. 'Rope' remains the name of the two-strand wound object, and the Rope Hypothesis credited to Bill Gaede keeps its name. Identifiers (repository, rope_solver, file names, claim IDs) are unchanged.

Addendum (v3.5.0): the constant G interrogated, and the lattice fork

The strong-field campaign of 2026 opened by auditing this paper's oldest habit: the identification G ~ 1/(T a), assumed since the sector's first derivation. The audit (GRV-074) found it DIMENSIONALLY OPEN — the same defect class the corpus had already met once elsewhere — while certifying that the sector's one live discriminating prediction is G-FORM-ROBUST: every candidate closure of the dimensions leaves the binary-pulsar refutability intact at millions of clock sigmas. Values were conditional; decidability was not. The repair (GRV-075) produced the corpus's first DERIVED G exponent pair, by the Sakharov route with the band coefficient recomputed on the sector's own machinery: G = c^3 a^2/(16 pi zeta hbar) with zeta = 1.208, tension-inert, both exponents fixed. Solving for the lattice scale SELECTS a = 1.26e-34 metres — eight Planck lengths — from a 96-site lattice zero-point coefficient that was never told about Planck. A survey of every route by which the corpus ever set its lattice scale (GRV-076) then found ZERO live pins: the familiar scale sets are the vacuum density bound wearing a different hat, every physical invariant rides the degeneracy untouched, and the apparent two-scale tension is really a single-bit FORK — is the Sakharov identification true? — with two named internal discriminators. The fork now matters at strong field: the Hawking-form coefficient of the black-hole paper's closing sections differs between its branches by eighteen orders of magnitude, which is the strongest internal lever the corpus owns for deciding it. [UPDATE 2026-08-09, GRV-101: the 96-site coefficient was EXTENSIVE -- the registered zeta carried the ring size sqrt(96), and the corrected intensive chain selects a = 0.80 Planck lengths, with the covariant chain giving (0, 0.18] l_P on the positivity half-line. The PLANCK-CLASS selection stands and is stronger; the digit eight does not.]